

Dual-Innovation Neural-Kalman Fusion for Real-Time Indoor Localization

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Abstract—Indoor positioning using hybrid smartphone sensors (Wi-Fi, Magnetic, and IMU) has seen growing interest, but deep-learning models often struggle to generalize in real-world deployments. This paper identifies key limitations in current regression methods, such as trajectory memorization, body-frame magnetic direction dependency, and non-causal look-ahead bias. To address these issues, we propose a generalized, decoupled, and strictly causal Neural-Kalman fusion framework for hybrid indoor localization. The architecture enforces a mathematical separation between spatial measurement extraction and temporal motion estimation. A Multi-Layer Perceptron (MLP) environment model maps dense Wi-Fi fingerprints from any N Access Points to a 2D probability heatmap over M discrete nodes. Simultaneously, a 1D Convolutional Neural Network (CNN) matches rotation-invariant magnetic sequences to structural anomaly maps. These spatial estimates are fused with a Pedestrian Dead Reckoning (PDR) motion model using an extended KalmanNet framework. The system replaces the fixed linear-Gaussian gain with dual 2×2 matrix gains dynamically produced by a Gated Recurrent Unit (GRU), utilizing availability masks to operate seamlessly even when only a subset of sensors is available. Evaluated rigorously on continuous real-world trajectories, the generalized system attains an error of 0.47 m under standard conditions and 1.07 m during degraded Wi-Fi scenarios, representing a 25.3% improvement over single-innovation baselines.

Index Terms—Indoor positioning, Extended Kalman Filter, KalmanNet, sensor fusion, pedestrian dead reckoning, hybrid sensors.

I. INTRODUCTION

Indoor localization is a critical technology for location-based services, smart buildings, and emergency navigation [1]. Because satellite signals attenuate severely indoors, systems must rely on ambient multi-modal smartphone sensors, primarily Wi-Fi Received Signal Strength Indication (RSSI), geomagnetic data, and Inertial Measurement Units (IMUs) [2], [3]. These signals are highly complementary: Wi-Fi offers absolute but infrequent position fixes, while IMU dead reckoning gives smooth, high-rate motion updates but suffers from cumulative drift [4]. The geomagnetic field, meanwhile, presents a ubiquitous and stable distortion pattern inside modern steel-reinforced buildings, providing a dense source of structural context [5].

The fundamental challenge in hybrid indoor localization is developing a unified model that optimally fuses these discrete

modalities while retaining robustness to sensor outages. Recent methodologies treat positioning as an end-to-end regression problem, utilizing Convolutional Neural Networks (CNNs) and Bidirectional Long Short-Term Memory networks (Bi-LSTMs) to map sequences directly to Cartesian coordinates [6]–[8]. Advanced topological approaches have also utilized Graph Neural Networks (GNNs) [9], and virtual mapping techniques have attempted to generalize across physical structures [10]. However, standard sequence models possess critical limitations in live environments: they frequently memorize specific spatial routes rather than generalizing the environment, they suffer from heading-bias if raw body-frame magnetic vectors are used, and bidirectional architectures are fundamentally non-causal.

To address these limitations, we propose a generalized, decoupled, strictly causal Neural-Kalman fusion architecture designed for hybrid localization. Our framework separates spatial environment learning from temporal tracking. We construct generic models for any number of Wi-Fi Access Points (N) and spatial reference nodes (M), allowing broad applicability. Spatial updates are fused with a classical Pedestrian Dead Reckoning (PDR) model using a dual-innovation KalmanNet [11], which calculates context-dependent trust matrices dynamically and natively handles missing sensor modalities via input availability masks.

The remainder of this paper is structured as follows. Section II mathematically formalizes the proposed generalized system architecture. Section III details the experimental setup, dataset processing, and augmentation. Section IV presents the performance evaluation, and Section V concludes the paper.

II. PROPOSED SYSTEM ARCHITECTURE

Our proposed architecture functions as a modular, learned Bayesian filter designed for general hybrid indoor environments. It relies entirely on past and present observations, maintaining strict causality. The complete system architecture is illustrated in Fig. 1.

A. General Problem Formulation

Assume a targeted indoor environment characterized by N uniquely identifiable Wi-Fi Access Points (APs) and a discrete reference grid containing M surveyed spatial nodes (Reference Points). At any discrete time step t , the smartphone yields a

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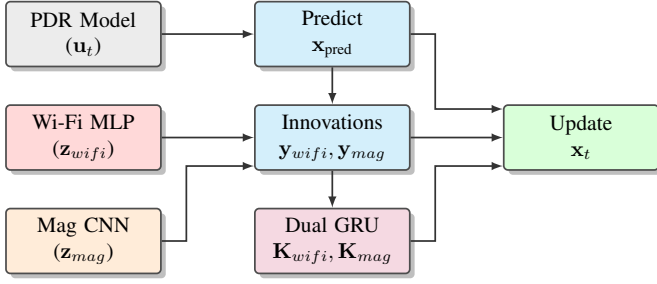


Fig. 1: The proposed Neural-Kalman Fusion Architecture. Both the Wi-Fi probability heatmap and the continuous 1D-CNN magnetic fix generate innovations against the PDR prediction. A recurrent Dual GRU observes these innovations to compute optimal Kalman gain matrices for the state update.

subset of sensory data: an IMU vector sequence (acceleration and gyroscope), a 3D magnetic field vector $\mathbf{m}_t \in \mathbb{R}^3$, and a sparsely sampled Wi-Fi RSSI vector $\mathbf{s}_t \in \mathbb{R}^N$. Our objective is to continuously estimate the user's 2D Cartesian spatial state $\mathbf{x}_t = [x_t, y_t]^T \in \mathbb{R}^2$.

B. Wi-Fi Processing and Heatmap Model

The raw Wi-Fi scan $\mathbf{s}_t \in \mathbb{R}^N$ is preprocessed into a normalized input $\tilde{\mathbf{s}}_t \in [0, 1]^N$. Absent APs, i.e., those that are not detected by the device, are assigned a constant floor value of -100 dBm. Each element is then clipped and linearly rescaled as

$$\tilde{s}_{t,i} = \begin{cases} 0 & \text{if } s_{t,i} \leq -100 \text{ (AP absent)} \\ \frac{\text{clip}(s_{t,i}, -90, -30) + 90}{60} & \text{otherwise} \end{cases} \quad (1)$$

To map this highly non-linear signal to spatial coordinates, we employ a Multi-Layer Perceptron (MLP) with Dropout regularization. The MLP processes $\tilde{\mathbf{s}}_t$ and outputs a probability vector $\mathbf{p} \in \mathbb{R}^M$ via a Softmax layer. Each element p_c represents the posterior probability that the device occupies grid cell c given the observed RSSI vector:

$$p_c = P(\mathbf{x}_t \in \text{cell } c \mid \tilde{\mathbf{s}}_t), \quad \sum_{c=1}^M p_c = 1 \quad (2)$$

The training target $\mathbf{q} \in \mathbb{R}^M$ is a discrete probability distribution constructed by evaluating a 2D isotropic Gaussian, centered at the true position $\mathbf{x}_{true}$, at each of the M grid cell coordinates $\mathbf{c}_c$, and normalizing the result to sum to unity:

$$q_c = \frac{\exp\left(-\frac{\|\mathbf{c}_c - \mathbf{x}_{true}\|^2}{2\sigma^2}\right)}{\sum_{j=1}^M \exp\left(-\frac{\|\mathbf{c}_j - \mathbf{x}_{true}\|^2}{2\sigma^2}\right)}, \quad \sum_{c=1}^M q_c = 1 \quad (3)$$

where σ is the spatial spread parameter. A Gaussian target is preferable to a one-hot label because it assigns partial probability mass to cells in the vicinity of the true location, providing the network with a smooth, spatially-graded gradient signal rather than a binary right-or-wrong penalty. This significantly

stabilizes training in regions of sparse survey coverage. The network is then optimized by minimizing the KL divergence between $\mathbf{q}$ and the network output $\mathbf{p}$:

$$\mathcal{L}_{MLP} = D_{KL}(\mathbf{q} \parallel \mathbf{p}) = \sum_{c=1}^M q_c \log\left(\frac{q_c}{p_c}\right) \quad (4)$$

During real-time inference, the continuous coordinate estimate $\mathbf{z}_{wifi}$ and the associated measurement covariance matrix $\mathbf{R}_{hm}$ are computed as the probability-weighted expectation (soft-argmax):

$$\mathbf{z}_{wifi} = \sum_{c=1}^M p_c \cdot \mathbf{c}_c \quad (5)$$

$$\mathbf{R}_{hm} = \sum_{c=1}^M p_c (\mathbf{c}_c - \mathbf{z}_{wifi})(\mathbf{c}_c - \mathbf{z}_{wifi})^T \quad (6)$$

where $\mathbf{c}_c \in \mathbb{R}^2$ denotes the physical coordinates of grid cell c . A concentrated heatmap yields a tightly bounded covariance, providing an honest, per-scan confidence estimate to the downstream filter.

C. Magnetic Sequence Matcher

To mitigate orientation bias inherent in raw $\mathbf{m}_t$ vectors, we extract 4 rotation-invariant scalar features per frame from the magnetic field vector and the estimated gravity direction $\hat{\mathbf{g}}_t$ (obtained from the low-pass filtered accelerometer):

$$\begin{aligned} m_N &= \|\mathbf{m}_t\|, & m_V &= \frac{\mathbf{m}_t \cdot \hat{\mathbf{g}}_t}{\|\hat{\mathbf{g}}_t\|} \\ m_H &= \|\mathbf{m}_t - m_V \hat{\mathbf{g}}_t\|, & \delta &= \arctan\left(\frac{m_V}{m_H}\right) \end{aligned} \quad (7)$$

where m_N is the field magnitude, m_V the vertical projection, m_H the horizontal component, and δ the magnetic dip angle. All four are scalar and invariant to device orientation. Because single-point magnetic matching is highly ambiguous, we use a 1D Convolutional Neural Network (CNN) to map a temporal sliding window of length T over these 4-channel features to a spatial coordinate.

The CNN processes the input to yield both a continuous position fix $\mathbf{z}_{mag} \in \mathbb{R}^2$ and a calibrated scalar log-variance $\log(\sigma_{mag}^2)$. It is optimized using a heteroscedastic Negative Log-Likelihood (NLL) loss, which balances coordinate accuracy against self-reported confidence:

$$\mathcal{L}_{CNN} = \frac{1}{B} \sum_{i=1}^B \left(\frac{\|\mathbf{z}_{mag}^{(i)} - \mathbf{z}_{true}^{(i)}\|^2}{2\sigma_{mag}^{2(i)}} + \frac{1}{2} \log(\sigma_{mag}^{2(i)}) \right) \quad (8)$$

where B is the batch size, $\mathbf{z}_{true}$ is the ground-truth coordinate, and the variance $\sigma_{mag}^2 = \exp(\logvar)$ is clamped to a minimum of 0.01 for numerical stability.

Because all four input features (m_N , m_V , m_H , δ) are rotation-invariant scalars derived from the field geometry rather than from the device body frame, the magnetic sequence matcher is inherently agnostic to device posture. The MagWi dataset covers three distinct holding modes—standard navigation hold, call-listening (device near the ear), and natural arm

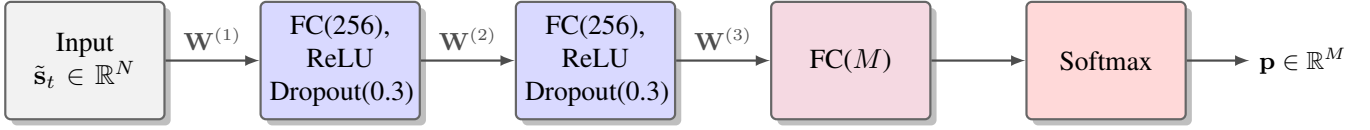


Fig. 2: Architecture of the Wi-Fi Multi-Layer Perceptron (MLP) environment model. Dense layers sequentially extract features from the RSSI vector to compute an M -dimensional probability heatmap.

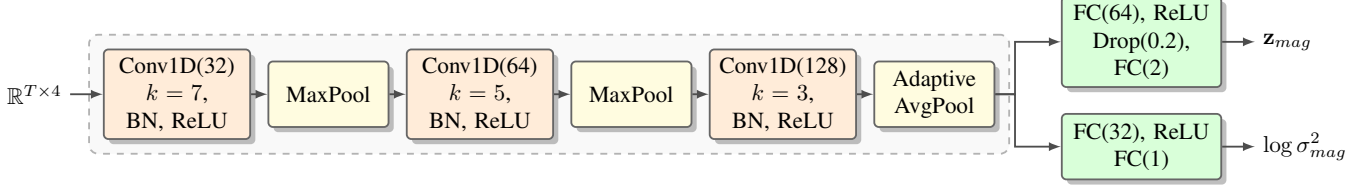


Fig. 3: Architecture of the 1D-CNN Magnetic Sequence Matcher. A shared convolutional encoder with BatchNorm extracts temporal features, which are processed by two independent heads: a position head (with Dropout) predicting a 2D coordinate, and a variance head estimating heteroscedastic uncertainty.

swinging—and no postural calibration or additional adaptation is required across any of them.

D. Causal Pedestrian Dead Reckoning (PDR)

The Pedestrian Dead Reckoning (PDR) system computes relative displacement at the full IMU rate ($f_s = 16.7$ Hz). Let $a_t = \|\mathbf{a}_t\| = \sqrt{a_x^2 + a_y^2 + a_z^2}$ denote the scalar acceleration magnitude. Step detection is performed by a first-order exponential moving average (EMA) low-pass filter that tracks the quasi-static gravitational baseline. The filter is initialized to $\bar{a}_0 = 9.81$ m/s² and updated each frame as:

$$\bar{a}_t = \alpha \bar{a}_{t-1} + (1 - \alpha) a_t, \quad \alpha = 0.98 \quad (9)$$

The high-pass residual $\tilde{a}_t$, which isolates dynamic foot-impact transients by removing the gravitational component, is then:

$$\tilde{a}_t = a_t - \bar{a}_t \quad (10)$$

A step is declared at frame t if and only if two conditions are simultaneously satisfied:

$$\tilde{a}_t > \tau \text{ and } (t - t_{\text{last}}) > \Delta_r \quad (11)$$

where $\tau = 0.6$ m/s² is the impact threshold and $\Delta_r = \lfloor 0.3 \cdot f_s \rfloor = 5$ frames is the refractory guard that suppresses double-counting within a single footfall.

Upon detection, the discrete control displacement vector $\mathbf{u}_t$ is computed as:

$$\mathbf{u}_t = \begin{bmatrix} L_s \cos(\theta_t + \phi_h) \\ L_s \sin(\theta_t + \phi_h) \end{bmatrix} \quad (12)$$

where θ_t is the device yaw from the orientation sensor (column `Orn_z`) at time t . The two scalar calibration parameters are computed once from the training walks and then frozen for the test device. The step length L_s is obtained by dividing the true path length by the number of detected steps and averaging across training walks:

$$L_s = \frac{1}{|\mathcal{W}_{\text{train}}|} \sum_{w \in \mathcal{W}_{\text{train}}} \frac{\ell_w}{n_w^{\text{steps}}} \quad (13)$$

The heading offset ϕ_h is the circular mean of the per-frame angular difference between the true trajectory direction and the device heading, computed via the four-quadrant inverse tangent:

$$\phi_h = \arctan 2 \left(\overline{\sin(\theta_{\text{true}} - \theta_t)}, \overline{\cos(\theta_{\text{true}} - \theta_t)} \right) \quad (14)$$

where $\theta_{\text{true},t} = \arctan 2(\Delta y_t, \Delta x_t)$ is the instantaneous ground-truth heading and the overline denotes the mean over all training frames. The complete geometry of a single PDR step is illustrated in Fig. 4.

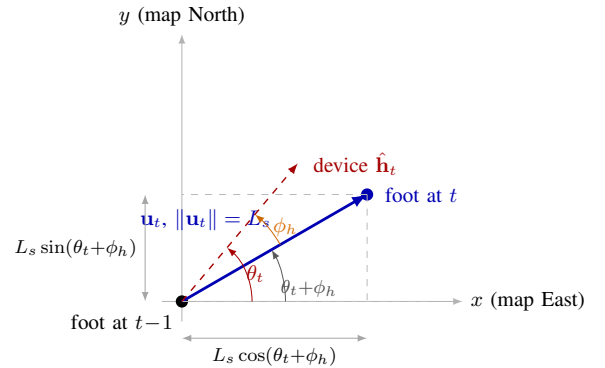


Fig. 4: Geometry of a single PDR step in the 2D map frame. The device orientation sensor reports heading θ_t ; a calibrated offset ϕ_h corrects the alignment between the device frame and the map frame. The resulting step displacement $\mathbf{u}_t$ has magnitude L_s and direction $\theta_t + \phi_h$, with Cartesian components as labeled.

E. Dual-Innovation KalmanNet Fusion

Classical EKFs rely on fixed covariance assumptions that fail to model highly variable multi-modal environments. We implement KalmanNet [11], replacing the analytical Kalman

gain with a learned one computed by a Gated Recurrent Unit (GRU). The prediction step follows standard dead reckoning:

$$\mathbf{x}_{\text{pred}} = \mathbf{x}_{t-1} + \mathbf{u}_t \quad (15)$$

The Wi-Fi innovation is a standard spatial residual:

$$\mathbf{y}_{wif i} = \mathbf{z}_{wif i} - \mathbf{x}_{\text{pred}} \in \mathbb{R}^2 \quad (16)$$

Next, the magnetic innovation, however, operates in the scalar anomaly domain rather than Cartesian space. The observed magnetic anomaly value at the current position is compared to the pre-surveyed map value $A(\mathbf{x}_{\text{pred}})$:

$$y_{\text{mag}} = A_{\text{obs}} - A(\mathbf{x}_{\text{pred}}) \in \mathbb{R} \quad (17)$$

To convert this scalar mismatch into a 2D spatial correction, it is scaled by the spatial gradient $\nabla A = [\frac{\partial A}{\partial x}, \frac{\partial A}{\partial y}]^T$ of the pre-surveyed anomaly field, computed via discrete finite differences on the reference grid.

The input feature vector to the GRU encompasses 13 variables: the Wi-Fi innovation $\mathbf{y}_{wif i} \in \mathbb{R}^2$, the scalar magnetic innovation $y_{\text{mag}} \in \mathbb{R}$, the field gradient $\nabla A \in \mathbb{R}^2$, the temporal difference of consecutive Wi-Fi fixes $\Delta \mathbf{z}_{wif i} \in \mathbb{R}^2$, the PDR control $\mathbf{u}_t \in \mathbb{R}^2$, the previous state update $\Delta \mathbf{x}_{t-1} \in \mathbb{R}^2$, and two binary availability masks $m_{wif i}, m_{\text{mag}} \in \{0, 1\}$. The GRU outputs an 8-dimensional vector, reshaped into two independent 2×2 gain matrices, $\mathbf{K}_{wif i}$ and $\mathbf{K}_{\text{mag}}$. The posterior state $\mathbf{x}_t$ is updated as:

$$\mathbf{x}_t = \mathbf{x}_{\text{pred}} + m_{wif i} \cdot \mathbf{K}_{wif i} \mathbf{y}_{wif i} + m_{\text{mag}} \cdot \mathbf{K}_{\text{mag}} (y_{\text{mag}} \cdot \nabla A) \quad (18)$$

This structural formulation ensures the system operates flexibly across any sensor sub-configuration. When Wi-Fi is unavailable, $m_{wif i} = 0$ nullifies the Wi-Fi correction term in (18), and the filter relies solely on the continuous magnetic gradient updates and PDR motion. Conversely, in magnetically featureless regions where $\nabla A \approx \mathbf{0}$, the GRU learns to suppress $\mathbf{K}_{\text{mag}}$ and wait for the next Wi-Fi fix.

III. EXPERIMENTAL SETUP AND DATASET

To stringently evaluate the proposed generalized architecture, we selected the MagWi Benchmark Dataset [12], specifically focusing on the IT Engineering building. This multi-floor structure provides a highly complex, dense pathway network ideal for testing tracking resilience. While validated here, the mathematical framework is entirely building-agnostic; it can be deployed on any indoor structure following an initial reference grid survey.

A. Dataset Augmentation

Continuous recordings within the dataset only label the starting node, lacking per-frame trajectory ground truth necessary for rigorous temporal evaluation. We generate synthetic but physically realistic trajectories to validate the KalmanNet.

- **Path generation:** An ε -graph is constructed over the true surveyed spatial grid nodes ($M = 168$). The ε -graph connects any two nodes that are physically separated by a distance $\leq \varepsilon$ (where $\varepsilon = 1.6$ m), yielding a connected,

topologically accurate corridor network. Trajectories traverse shortest paths between randomized node endpoints.

- **Sensor simulation:** Gait speed is sampled uniformly between 1.0 and 1.35 m/s. The simulated heading θ_t tracks the true geometric path tangent, subjected to a random-walk drift and an 8.8° white noise variance to identically match the empirically measured gyroscope noise characteristics of the dataset.

Using this robust pipeline, we generated 250 trajectories for training and 60 independent trajectories for testing.

B. Training Details

All neural network components were developed using the PyTorch framework. The MLP and CNN were optimized via Adam ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}) with an initial learning rate of 10^{-3} , dynamically decayed via a ReduceLROnPlateau scheduler (patience 8, factor 0.5). The MLP uses a Dropout rate of $p = 0.3$ and a Gaussian target with spatial standard deviation $\sigma = 2.0$ m. The MLP was trained for 80 epochs; the CNN for 60 epochs. The 1D-CNN magnetic sequence matcher utilized an empirically optimal temporal window of $T = 84$ frames (5.0 s at 16.7 Hz), selected via a sweep over candidate $\{50, 84, 134, 167\}$ frames, providing sufficient spatial variation for unambiguous matching without overfitting long topological routes. The KalmanNet GRU was trained for 150 epochs via Adam with a learning rate of 2×10^{-3} , weight decay of 10^{-5} , and MSE loss against ground-truth trajectories.

To stringently evaluate device generalization, data originating from the Samsung Galaxy S9+ was explicitly held out during the training phase and reserved solely for evaluation.

IV. EXPERIMENTAL RESULTS

A. Environment Model Accuracy

The standalone performance of the Wi-Fi heatmap environment model prior to temporal fusion is shown in Table I. The MLP achieves a robust MAE of 1.43 m on standard splits. Crucially, it maintains a high degree of accuracy of 2.02 m MAE on entirely unseen smartphone hardware.

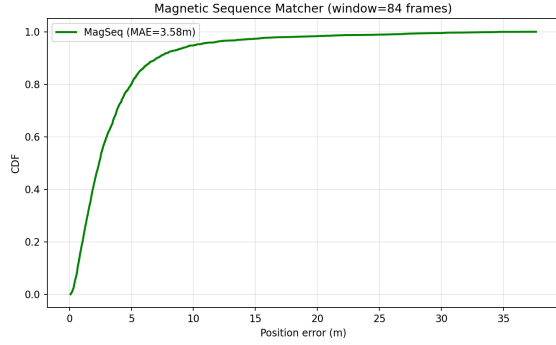
TABLE I: Wi-Fi Heatmap Environment Model Performance

Configuration	Test Device	MAE	Med.	P90	Max
Random split	Mixed	1.43 m	1.12 m	2.65 m	7.84 m
Held-out device	Samsung S9+	2.02 m	1.68 m	3.74 m	9.12 m

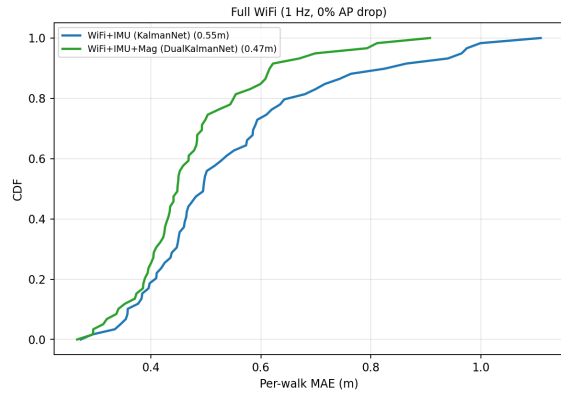
Figure 5 (top) displays the continuous spatial matching error CDF for the 1D-CNN magnetic sequence model (MAE of 3.58 m), validating its capability as an effective high-rate structural anchor during periods of Wi-Fi attenuation.

B. Dual-Innovation KalmanNet Fusion Comparison

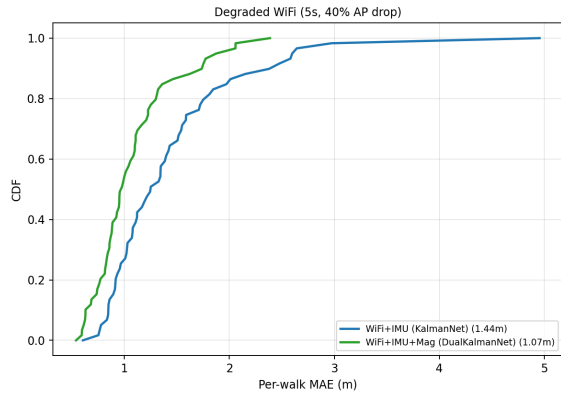
Table II details the comparative performance of the temporal fusion mechanisms across the independent test walks, evaluated under two separate signal availability regimes.



(a) Standalone 1D-CNN magnetic sequence matcher (MAE 3.58 m).



(b) Temporal fusion under full Wi-Fi conditions (1 Hz).



(c) Temporal fusion under degraded Wi-Fi (5 s gap, 40% AP dropout).

Fig. 5: Cumulative Distribution Functions (CDF) of the tracking models. Top: standalone 1D-CNN magnetic sequence matcher error. Middle: Temporal fusion performance under full Wi-Fi conditions (1 Hz). Bottom: Temporal fusion performance under degraded Wi-Fi scenarios (5s gap, 40% AP dropout), highlighting the significant impact of the magnetic innovation.

TABLE II: Dual-Innovation Fusion Comparison (60 Test Walks)

Regime	Model Configuration	Mean Err.	Med.
Full Wi-Fi (1 Hz)	WiFi-only KalmanNet	0.55 ± 0.05 m	0.50 m
	DualKalmanNet (+Mag)	0.47 ± 0.03 m	0.45 m
Degraded Wi-Fi (5s gap, 40% Drop)	WiFi-only KalmanNet	1.44 ± 0.18 m	1.25 m
	DualKalmanNet (+Mag)	1.07 ± 0.10 m	0.97 m

In standard operating conditions featuring high-frequency 1 Hz Wi-Fi fixes, integrating the continuous magnetic sequence data provides a noticeable 13.4% improvement in localization accuracy. However, the true strength of the architecture is definitively revealed in the degraded signal regime (Figure 5, bottom). When Wi-Fi scans are artificially restricted to one reading every 5 seconds accompanied by a severe 40% missing AP dropout rate, the classical single-innovation approach drifts significantly. In contrast, the continuous 16.7 Hz magnetic updates act as an effective, dense bridge, yielding a substantial 25.3% reduction in Mean Absolute Error (improving from 1.44 m to 1.07 m).

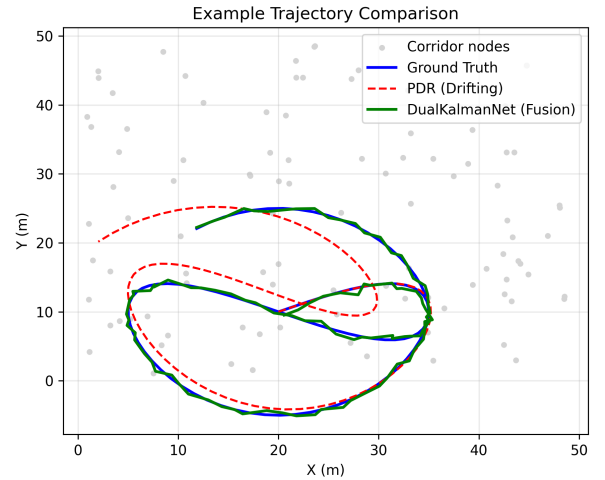


Fig. 6: Example trajectory visualization illustrating tracking divergence. Without frequent Wi-Fi spatial anchors, the standard PDR path (dashed red) drifts due to heading accumulation errors. The Dual-Innovation KalmanNet (solid green) tightly tracks the ground truth (solid blue) by integrating structural magnetic sequence measurements continuously.

Fig. 6 shows a representative test walk under the degraded Wi-Fi regime. The PDR-only trajectory (dashed red) diverges progressively from the ground truth (solid blue) as heading errors accumulate during the inter-scan gaps; the divergence is most visible in corridor turns where a small heading bias compounds over multiple steps. The Dual-Innovation KalmanNet (solid green), receiving scalar magnetic anomaly corrections at 16.7 Hz throughout each outage, suppresses this lateral drift and maintains close alignment with the true path. Upon each Wi-Fi fix arrival, the filter rapidly re-anchors, preventing

further error accumulation.

V. CONCLUSION

This paper introduced a generalized, decoupled Neural-Kalman fusion architecture designed to resolve pervasive limitations in deep-learning indoor positioning systems across Hybrid Wi-Fi, Magnetic, and IMU data regimes. By strictly separating spatial inference from temporal tracking modules, the framework successfully circumvents trajectory memorization and orientation-dependent data bias. The integration of a 1D-CNN magnetic sequence matcher alongside an MLP Wi-Fi probability heatmap enables robust spatial updates. Fusing these estimates through an extended KalmanNet architecture enables the system to continuously calculate context-dependent 2×2 matrix gains, leveraging availability masks to maintain resilience against missing sensor modalities.

In challenging, degraded scenarios—a common reality in physical deployments—the dual-innovation system effectively prevents inertial drift, delivering an impressive 25.3% performance increase over standard models. While current configurations necessitate building-specific training due to unique foundational AP signatures, the mathematical structure is universally applicable. Future research efforts will focus on expanding the architecture to facilitate full three-dimensional, multi-floor positioning by integrating barometric pressure altitude data into the KalmanNet state vector.

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